

# Household Identity Walkthrough

A CTV/OTT identity graph analysed end to end – business requirements in, ranked findings out – shown against the live AdTech Demo workspace.

Workspace AdTech Demo / Household Identity      Graph AdtechGraph

Database addtech-knowledge-graph

## What you are showing

It turns a business requirements document into evidenced graph findings. Nobody writes a query, picks an algorithm, or describes the schema – the system reads the database, works out what to ask, runs it, and explains the answer in the language of the industry.

Six specialist agents run in sequence. Each one's output is recorded, so a finding can be traced back through the analysis that produced it to the sentence in the requirements that asked for it.

|                     |                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SchemaAnalyst       | Reads the live database and works out what the graph actually contains – collections, edge directions, which entities are core and which are reference data.        |
| RequirementsAnalyst | Extracts objectives, constraints and success criteria from the customer's own document.                                                                             |
| UseCaseExpert       | Proposes analyses that would answer those objectives against this particular schema.                                                                                |
| TemplateEngineer    | Chooses the algorithm and the collections it should run over. Algorithm-aware: connectivity analyses exclude reference collections that would distort them.         |
| ExecutionSpecialist | Runs the work on ArangoDB's Graph Analytics Engine – compute spun up for the analysis and torn down after it.                                                       |
| ReportingSpecialist | Turns ranked vertices into findings with confidence scores, business impact and prioritised recommendations, phrased for the industry it was told it is working in. |

# Why this matters in ad-tech

The domain problem, so the findings in beats 5 and 6 land as answers rather than as statistics.

A household watches CTV on a television, browses on a phone, and converts on a laptop. If you cannot tell those are one household, you over-count reach, mis-attribute the conversion, and pay a third party — LiveRamp, Cadent — for the linkage. The objective in this workspace is to derive that linkage in-house: stitch roughly **330 million devices** to **120 million residential IPs** into stable household clusters.

The graph is built from no-bid logs — the bid requests the platform saw and did not win. Free observational exhaust, already in the building.

| ENTITIES · 9 COLLECTIONS                                                                         | RELATIONSHIPS · 7 COLLECTIONS                                                                   |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| AppProduct · Device · Exchange · IP ·<br>InstalledApp · Location · Publisher ·<br>Site · SiteUse | INSTANCE_OF · LOCATED_IN · OWNED_BY ·<br>SEEN_ON_APP · SEEN_ON_IP · SEEN_ON_SITE<br>· SERVED_BY |

Three algorithms carry the demo. What each one *means* here is the part worth saying out loud:

- PageRank

Which nodes are gateways. Applied to exchanges and publishers it finds the few through which most inventory flows — where data-quality SLAs and fraud monitoring earn the most.
- SCC

Whether relationships are reciprocal. A device and an IP that point back at each other are evidence of a household. One-way links are not. This is a precondition for clustering, which is why beat 6 matters.
- WCC

Household candidates. The ad-tech vertical carries the domain's own thresholds: 3–18 devices is a plausible household, a cluster holding over 40% of nodes is over-aggregation from a public IP, singletons mean fragmentation, and more than 20 devices across more than 10 IPs is a botnet signature rather than a family.

Those thresholds are why the reporting reads as ad-tech rather than as graph theory. The same engine primed with a different vertical describes the same algorithms as financial-crime or corporate-disclosure findings.

## Before you present

Four checks. The last one decides whether you run an analysis live or stay on stored results — settle it before you are in front of anyone.

☐ **Both services are up.**

API on :8010 , UI on :3000 . The API must start with `ARANGO_DATABASE=aga_workspace` — that is the product's own metadata store, not the analytics graph. Started against the wrong database it comes up empty.

☐ **The banner reads `LIVE` .**

Top centre. If it says `ERROR` , the workspace failed to load — the panel will be empty rather than showing stand-in data, and it retries on its own within about 15 seconds.

☐ **Ten reports are listed, each ending `• AdtechGraph in addtech-knowledge-graph` .**

That suffix is what lets you answer "which database is this?" without leaving the screen.

☐ **Decide: live analysis, or stored results only?**

The Graph Analytics Engine returned `503` on every attempt during recent testing, while a single isolated run succeeded. If you have not proved it working *that morning*, present stored results and say so — the reports are real output from a real run either way.

# The walkthrough

Six beats. The through-line: a business document becomes a ranked, evidenced answer, and every artefact on screen says where it came from.

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## 1

0:00

### Open on the workspace, not the login

```
http://127.0.0.1:3000/workspace?workspaceId=workspace-adtech-demo
```

*"This is one customer project. Everything it touches is on the left – the database connection, the discovered graph, the requirements document, the run, and every report that run produced."*

Let them read the sidebar for a beat. Each row names what it is *and* where it points, which is the thing people usually have to ask about.

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## 2

1:00

### Start from the business document

```
Click AdTech_business_requirements.md
```

*"We start where the customer starts – a requirements document. Not a schema, not a query. The objective here is identity resolution autonomy: stitch roughly 330 million devices to 120 million residential IPs into stable household clusters, and stop paying third parties for it."*

Then click `Requirements`. It reads `v1 (approved) · from AdTech_business_requirements.md` — the version is attributed to the document it was extracted from, so provenance is visible rather than assumed.

### 3

3:00

#### Show the graph it reasoned over

Click [AdtechGraph](#) · Graph profile

*"The schema was discovered, not hand-written. Nine vertex collections, seven edge collections – devices, IPs, publishers, exchanges, apps, sites – and roughly 418,000 vertices across 543,000 edges. This is a real graph, not a fixture."*

| VERTICES | EDGES   | VERTEX COLLECTIONS | EDGE COLLECTIONS |
|----------|---------|--------------------|------------------|
| 417,888  | 542,889 | 9                  | 7                |

### 4

5:00

#### The run, as six agents

Click [Run run-09eb83bf...](#)

*"One run, six specialist agents in sequence: schema analysis, requirements extraction, use-case generation, template generation, execution on the Graph Analytics Engine, then reporting. Each step is recorded with its status – this is an audit trail, not a progress bar."*

All six read **completed**. Right-click a step for its artifacts if someone asks what a stage actually produced.

### 5

7:00

#### The finding worth stopping on

Click [Analysis Report: UC-S04: Ad-Exchange Gateway Analysis](#)

*"PageRank over 9,449 nodes. The top five exchanges carry 21.79% of the network's authority – five gateways that most inventory traffic flows through. That is where you put data-quality SLAs and fraud monitoring, and it is not something you would have guessed from the schema."*

Scroll to the interactive chart. Every report carries its summary, its insights with confidence scores, its recommendations with priority and expected impact, and **Export HTML** / **Export Markdown** in the header.

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## 6

9:30

### Close on the one that says "not yet"

Click Analysis Report: UC-S05: Device-IP Reciprocity Study

*"Strongly connected components found a 100% singleton rate – no bidirectional loops between devices and IPs at all. For household identity that is a finding, not a failure: the graph is currently one-way, so reciprocity has to be built in before 330 million devices will consolidate into households. The system said the thing the customer did not want to hear."*

This is the strongest close available. It shows the analysis is not flattering the data — and it maps straight back to the objective opened with in beat 2.

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### Do not click these

Each is empty, slow, or unfinished. None will crash — they will just cost you the room.

- **Analysis Catalog and Use Cases & Templates**

Both render honest empty states. Nothing was ever populated behind them.

- **Discover Graph on the `risk-intelligence` connection**

Discovery on a large database can run for minutes. One attempt against a 98-collection, 31.9-million-document database passed five minutes without returning.

- **Switching the active graph to `DataGraph`**

It is a valid profile on a second database, so switching *works* — but there are no runs or reports behind it. Mention that multi-database is supported; do not demonstrate it by switching.

### If something goes wrong

Lines that keep the room with you instead of watching you debug.

#### The banner turns red

The API dropped. It retries automatically — the panel goes empty rather than showing fake data, which is deliberate. *"That's the backend restarting — it recovers on its own in a few seconds."* Talk over it; do not reload.

## A live analysis fails

The engine is a shared cluster resource and has been refusing connections under load. *"The compute engine is busy – here's the same analysis from the run we did earlier."* Then open the stored report. This is why beat 5 uses a stored result by default.

## Someone asks about a UUID

Run and report identifiers are visible and ugly. *"Those are stable identifiers – every report traces back to the run and the graph that produced it, which is what makes the output auditable."* Then point at the `AdtechGraph` in `addtech-knowledge-graph` suffix.

## Someone asks whether it can do their data

The honest answer is the connect flow. *"You point it at a database, it discovers the schema itself, and you hand it a requirements document. That's the whole setup."* Show **Connect to Database** — it lists the databases on the cluster with no credentials typed — then cancel out.

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Every figure on this page was read from the live workspace, not estimated: vertex and edge counts from the AdtechGraph profile, result counts and findings from the stored report records.

Rehearse the click path once end to end before presenting. The beats assume stored results throughout; nothing here requires the Graph Analytics Engine to be reachable.